

Chapter 7d: Variational Methods and Optimization

Rate of Convergence and Fixed Point Theory

Based on Pathak's "An Introduction to Nonlinear Analysis and Fixed Point Theory"

Nonlinear Analysis and Fixed Point Theory

July 14, 2026

Outline

- 1 Variational Formulation for Linear and Nonlinear Problems
- 2 Mountain Pass Lemma and Critical Points
- 3 Variational Principles for Non-self-adjoint Equations
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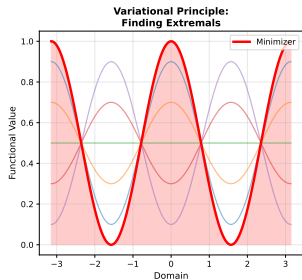
Overview: Variational Methods

- **Core Idea:** Transform differential equations into optimization problems
- **Advantage:** Direct connection between physical/mathematical problems and optimization
- **Framework:** Find extremal points of functionals in appropriate function spaces
- **Key Tools:**
 - Variational calculus
 - Functional analysis
 - Critical point theory

Definition

A **functional** $J : V \rightarrow \mathbb{R}$ maps functions in a Banach space V to real numbers.

Variational Principle: From Calculus of Variations



Variational Problem:

$$\min_{u \in V} J(u) = \int_D F(u, \nabla u) dV$$

Leads to Euler Equation:

$$\frac{\partial \mathcal{L}}{\partial u} - \nabla \cdot \frac{\partial \mathcal{L}}{\partial \nabla u} = 0$$

Critical Point = Solution

Classical Problem: Find the minimum of

$$J(u) = \int_D F(u, \nabla u) dV$$

subject to boundary conditions on ∂D .

Solution Method:

- 1 Compute the first variation: $\delta J = 0$
- 2 Derive Euler-Lagrange equation
- 3 Solve for critical points

Variational Problems: Concrete Example

Problem: Find extremals of the functional

$$I[y(x)] = \int x \sqrt{1 + y'^2} dx$$

Solution Steps:

- 1 Here $F(x, y, y') = x \sqrt{1 + y'^2}$
- 2 Euler equation: $F_y - \frac{d}{dx} F_{y'} = 0$
- 3 Compute: $F_y = 0$, $F_{y'} = \frac{xy'}{\sqrt{1+y'^2}}$
- 4 This gives: $\frac{d}{dx} \left(\frac{xy'}{\sqrt{1+y'^2}} \right) = 0$

Result: The extremals are parabolas: $(y - c_1)^2 = 4c_2(x - c_2)$

Symmetry and Self-Adjoint Operators

Key Observation: A variational principle exists if and only if the operator is self-adjoint.

For an operator $P : X \rightarrow X^*$ (where X^* is the dual space), the **symmetry condition** is:

$$\int \Psi P'_u \phi \, dV = \int \phi P'_u \Psi \, dV$$

Theorem 7.18 (Vainberg)

Suppose:

- 1 $P : X \rightarrow X^*$ is an operator from normed space X to its conjugate
- 2 P has a linear Gâteaux differential at every point
- 3 The bilinear functional $(DP(x, h_1), h_2)$ is continuous in x

Then P is potential (i.e., $P = \nabla F$) iff the symmetry condition holds.

Functional Formulation of Differential Equations

Problem: Given a differential equation $P(u) = f$ in domain D

Variational Reformulation:

$$\min_{u \in V} J(u) = \int_D [P(u) - f]^2 dV$$

Example: For Poisson equation $-\Delta u = f$:

$$J(u) = \int_D [\|\nabla u\|^2 - 2fu] dV \quad (\text{Dirichlet form})$$

Least Squares Method

For non-self-adjoint operators, the least squares variational formulation is:

$$J(u) = \int_D (Lu - f)^2 dV$$

which minimizes the L^2 residual.

Gâteaux and Fréchet Derivatives

Gâteaux Derivative: For a functional $F : X \rightarrow \mathbb{R}$:

$$F'_u(h) = \lim_{\varepsilon \rightarrow 0} \frac{F(u + \varepsilon h) - F(u)}{\varepsilon}$$

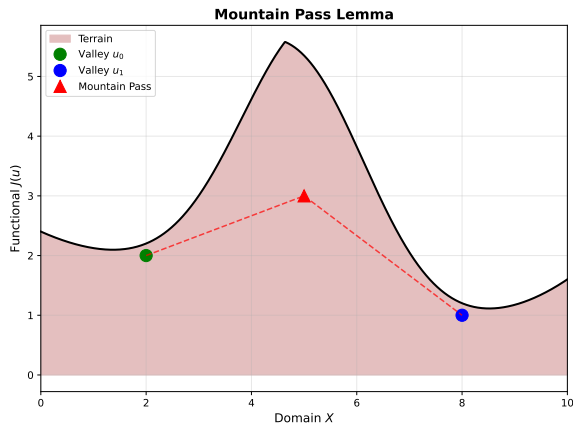
directional derivative in direction h

Fréchet Derivative: A continuous linear operator $A \in \mathcal{L}(X, Y)$ such that:

$$\lim_{h \rightarrow 0} \frac{\|F(x + h) - F(x) - Ah\|}{\|h\|} = 0$$

- **Fréchet** $\Rightarrow$ **Gâteaux** (always)
- **Gâteaux** $\Rightarrow$ **Fréchet** (under continuity conditions)

Mountain Pass Lemma (MPL)



Intuition: To go from one valley to another, we must cross a mountain pass.

Formal Statement: Let $J : X \rightarrow \mathbb{R}$ be a C^1 functional. Suppose:

- ① $\|u_1 - u_0\| > R_1$
- ② $J(u_0), J(u_1) < c_0 \leq \inf_{\|v - u_0\| = R_1} J(v)$

Then J has a critical value $c \geq c_0$ defined by:

$$c = \inf_p \max_{u \in p} J(u)$$

Palais-Smale Condition

Problem with MPL: The conclusion doesn't always hold!

Counterexample: $F(z) = |e^z - 1|^4$ on $\mathbb{C}$

- Minima at $z = 0$ and $2\pi i$
- Only critical value is at 0
- The mountain at πi exists but is not critical

Palais-Smale (PS) Condition

Any sequence $\{u_i\}$ in X for which:

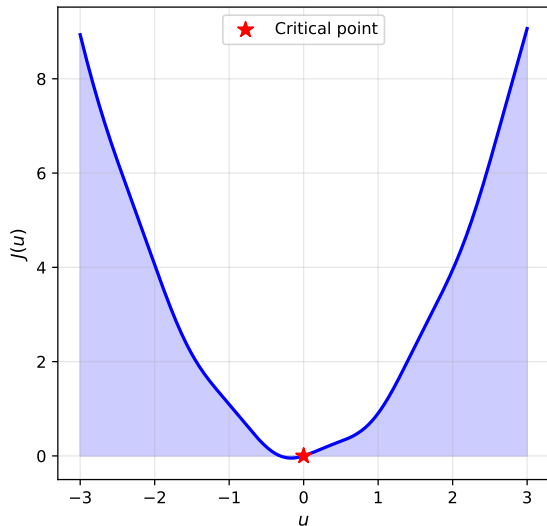
- 1 $J(u_i) \rightarrow c$ (bounded)
- 2 $J'(u_i) \rightarrow 0$ (derivatives vanish)

has a strongly convergent subsequence.

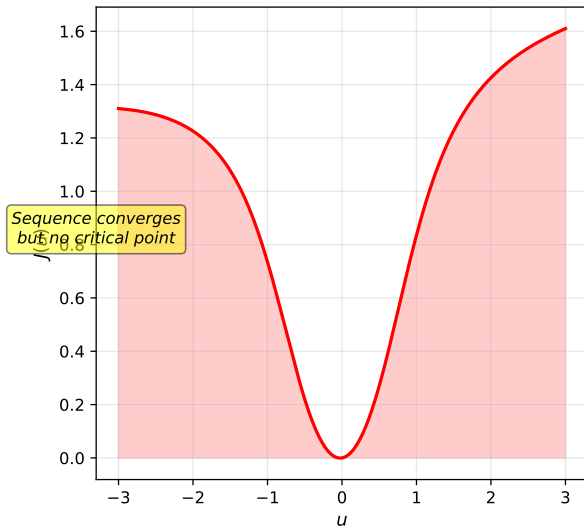
Conclusion: With PS condition, MPL guarantees a critical point at the mountain pass!

Example: Finding Critical Points

Satisfies Palais-Smale (PS)



Violates Palais-Smale (PS)



Adjoint Operators

Definition: For operator $A : \mathcal{H} \rightarrow \mathcal{H}$, the **adjoint** A^* satisfies:

$$(Ax, y) = (x, A^*y) \quad \forall x, y \in \mathcal{H}$$

Self-Adjoint: $A = A^*$

Problem: Non-self-adjoint equations don't admit variational principles directly!

Solution: Transform via least squares:

$$J(u) = \int_V (Lu - f)^2 dV$$

This is always variational, even for non-self-adjoint L .

Key Insight

The minimum of $J(u) = \int_V (Lu - f)^2 dV$ gives a least-squares solution, even when L is not self-adjoint.

Boundary Value Problems

General Boundary Value Problem:

$$Lu = f \quad \text{in } V \quad (1)$$

$$B_i u = 0 \quad \text{on } S \quad (2)$$

where S is the piecewise smooth boundary of domain V .

When L is self-adjoint:

$$J(u) = \frac{1}{2} \int_V uLu \, dV \quad \Rightarrow \quad \min J \text{ gives solution}$$

When L is non-self-adjoint:

$$J(u) = \int_V (Lu - f)^2 dV \quad \Rightarrow \quad \min J \text{ gives least-squares solution}$$

Hamiltonian Formulation

Classical Problem: Second-order equation $T^*Tq = f$

Hamiltonian Reformulation: Split into first-order system:

$$Tx = \rho \quad (3)$$

$$T^*\rho = f \quad (4)$$

where ρ represents generalized momentum.

Hilbert Space Formulation: Let $\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_2$, then:

$$L(x, Tx) = (Tx, \tilde{p}) - \frac{1}{2}(\tilde{p}, \tilde{p}) - (f, x)$$

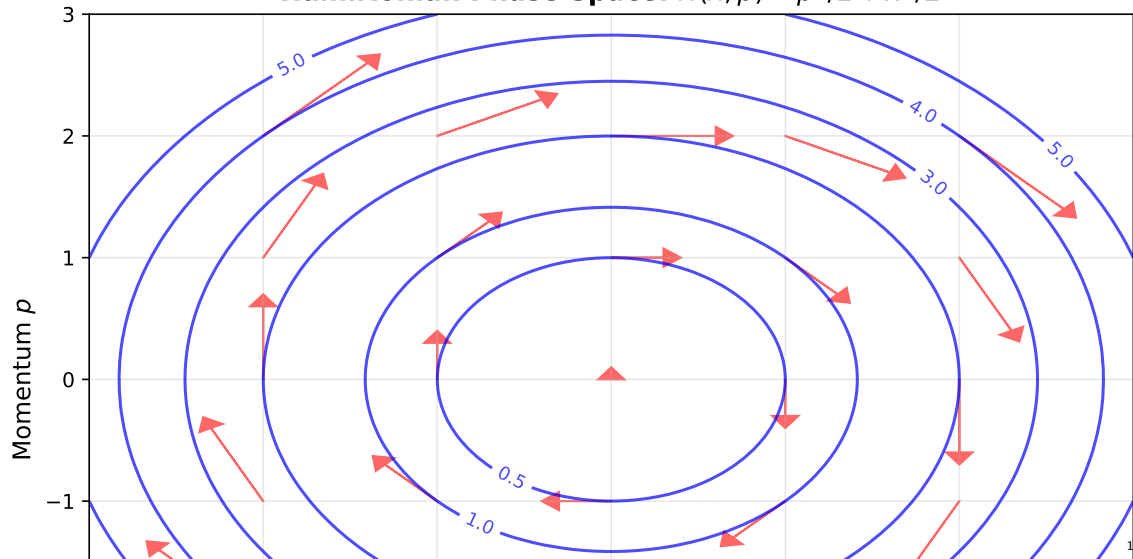
Hamilton's Canonical Equations

$$Tx = W_p, \quad T^*p = W_x$$

where $W(x, p) = \frac{1}{2}(p, p) + (f, x)$

Hamiltonian Phase Space Dynamics

Hamiltonian Phase Space: $H(x, p) = p^2/2 + x^2/2$



Lagrangian and Hamiltonian Duality

Legendre Transformation: The duality between Lagrangian and Hamiltonian:

$$\mathcal{H} = L - (Aq, p)$$

where $p = \frac{\partial L}{\partial(Aq)}$

Connection:

- Lagrangian: $\min J = \int L(q, \dot{q}) dt$
- Hamiltonian: Reformulate using conjugate momentum p
- Both describe same dynamics: $Tq = p(t)$, $T^*p = f(q(t), t)$

Mechanics Analogy

- x = generalized displacement
- $p = T^*x' =$ generalized momentum
- $\mathcal{H} = \frac{1}{2}(p, p) + V(x) =$ total energy

Schrödinger Equation from Variational Principle

Problem: Find wave function Ψ that extremizes:

$$\int_D \Psi^* (\mathcal{H}\Psi) dx dy dz$$

subject to normalization $\int_D \Psi^* \Psi dx dy dz = 1$

where $\mathcal{H} = -k\nabla^2 + V(x, y, z)$ is the Hamiltonian operator.

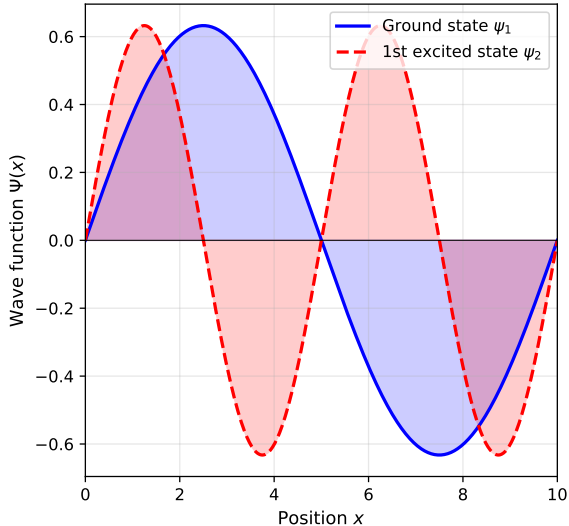
Solution Method:

- 1 Introduce Lagrange multiplier λ for constraint
- 2 Compute Euler equations
- 3 Result: $-k\nabla^2\Psi + V\Psi = \lambda\Psi$

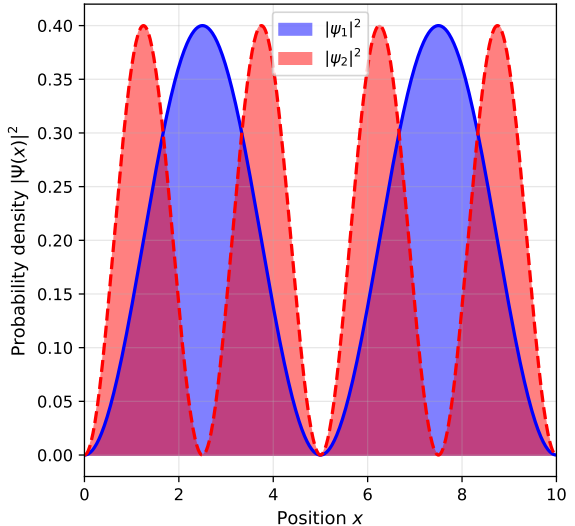
Conclusion: The time-independent Schrödinger equation is an eigenvalue problem!

Schrödinger Equation Example

Wave Functions: Particle in a Box



Probability Densities



Klein-Gordon Equation

Relativistic Wave Equation:

$$\nabla^2 \Psi - \frac{1}{c^2} \frac{\partial^2 \Psi}{\partial t^2} - \left(\frac{mc}{\hbar} \right)^2 \Psi = 0$$

Variational Formulation: Lagrangian density:

$$L = -\frac{\hbar^2}{2m} \left[\nabla \Psi^* \cdot \nabla \Psi - \frac{1}{c^2} \left(\frac{\partial \Psi}{\partial t} \right)^2 + \left(\frac{mc}{\hbar} \right)^2 \Psi^* \Psi \right]$$

- Minimizing action $\int L dt dx$ yields Klein-Gordon equation
- Extends Schrödinger to relativistic domain
- Predicts antiparticles (hole theory)

Fixed Point Problems

Abstract Fixed Point Problem: Find $x^* \in X$ such that:

$$x^* = T(x^*)$$

where $T : X \rightarrow X$ is a mapping on Banach space X .

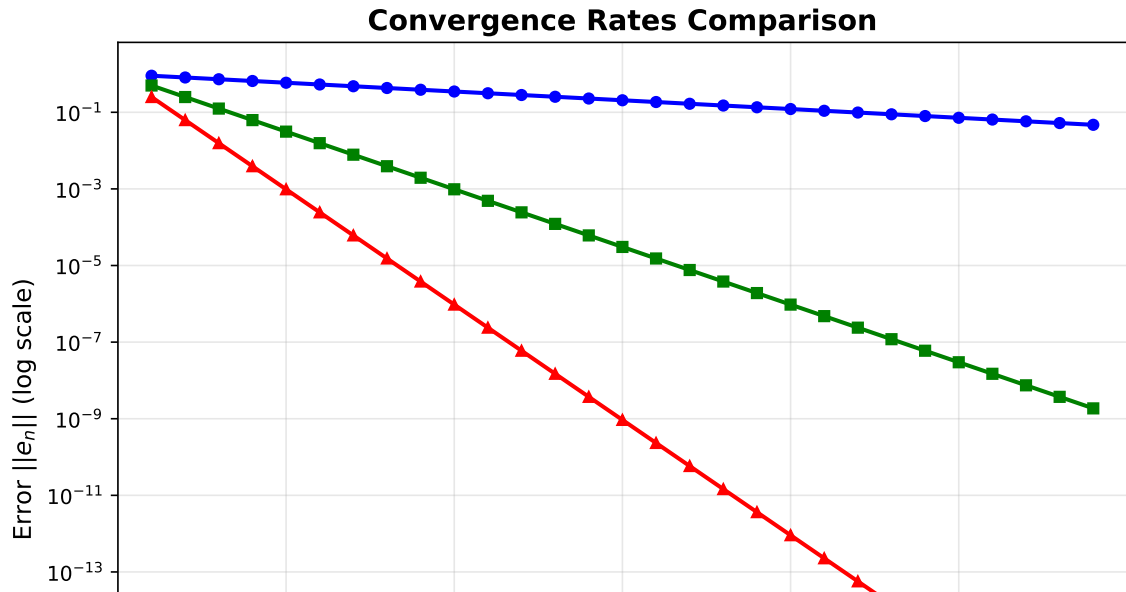
Iterative Solution:

$$x_{n+1} = T(x_n)$$

Fundamental Questions

- 1 Does $\{x_n\}$ converge to x^* ?
- 2 How fast does it converge?
- 3 Can we characterize the convergence rate?

Convergence Rates



Convergence Rate Definitions

Error: $e_n = \|x_n - x^*\|$

Linear Convergence:

$$\limsup_{n \rightarrow \infty} \frac{e_{n+1}}{e_n} = r < 1$$

- $e_n \sim C \cdot r^n$ for some constant C
- Slowest rate (exponential decay)

Superlinear Convergence:

$$\lim_{n \rightarrow \infty} \frac{e_{n+1}}{e_n} = 0$$

Quadratic Convergence:

$$\limsup_{n \rightarrow \infty} \frac{e_{n+1}}{e_n^2} = C < \infty$$

- Exponential improvement: errors square at each step
- Fastest practical rate

Contraction Mapping Theorem and Convergence

Banach Contraction Mapping Theorem

If $T : X \rightarrow X$ is a contraction mapping with constant $L < 1$:

$$\|T(x) - T(y)\| \leq L \|x - y\| \quad \forall x, y \in X$$

Then there exists unique fixed point $x^* = T(x^*)$ and:

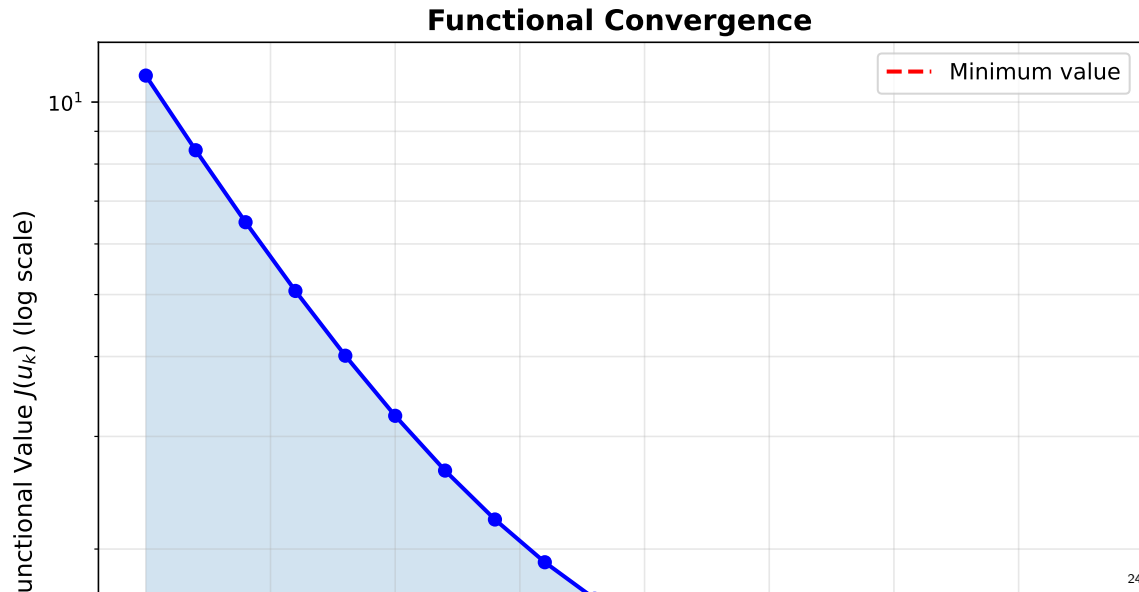
$$\|x_n - x^*\| \leq \frac{L^n}{1 - L} \|x_1 - x_0\|$$

Conclusion: Contraction mappings guarantee **linear convergence** with rate $r = L$.

Faster Rates: For non-contractive mappings (like Newton's method):

- Requires stronger conditions (e.g., differentiability)
- Can achieve quadratic or superlinear convergence

Functional Evolution During Convergence



Example: Euler-Lagrange Equation

Problem: Find extremals of the functional

$$I[y(x)] = \int_0^1 (y'^2 + y^2 + 1) dx, \quad y(0) = 1, \quad y(1) = 2$$

Solution Process:

- 1 Identify: $F(x, y, y') = y'^2 + y^2 + 1$
- 2 Compute: $F_y = 2y, \quad F_{y'} = 2y'$
- 3 Euler equation: $2y - \frac{d}{dx}(2y') = 0 \Rightarrow y'' - y = 0$
- 4 General solution: $y = c_1 e^x + c_2 e^{-x}$
- 5 Apply BC: $c_1 + c_2 = 1, \quad c_1 e + c_2 e^{-1} = 2$

Result: Explicit formula for extremal on interval $[0, 1]$

Example: Minimum Surface Area (Catenary)

Problem: Curve of revolution with minimum surface area connecting two circles.

Surface Area Functional:

$$A = \int_{x_1}^{x_2} 2\pi x \sqrt{1 + y'^2(x)} dx$$

Euler Equation: For $F = x\sqrt{1 + y'^2}$:

$$\frac{d}{dx} \left(\frac{xy'}{\sqrt{1 + y'^2}} \right) = \sqrt{1 + y'^2}$$

Solution: The catenary curve

$$y(x) = c_1 \cosh^{-1} \left(\frac{x}{c_1} \right) + c_2$$

Physical Application: Shape of hanging cable under gravity!

Rayleigh-Ritz Method for Eigenvalues

Problem: Approximate the first eigenvalue of $\nabla^2 z + \lambda z = 0$ in circle of radius 1.

Rayleigh-Ritz Approach:

- 1 Choose ansatz: $z(x, y) = (1 - r^2)$
- 2 Compute Rayleigh quotient: $\lambda \approx \frac{\int_D (\nabla z)^2 dA}{\int_D z^2 dA}$
- 3 Result: $\lambda_1 \approx 10.02$ (exact: ≈ 10.17)

Algorithm Structure:

```
def rayleigh_ritz(domain, ansatz_basis, bvp_params):  
    # Compute stiffness matrix K  
    K = assemble(grad(phi) * grad(phi))  
  
    # Compute mass matrix M  
    M = assemble(phi * phi)  
  
    # Solve generalized eigenvalue problem  
    eigvals, eigvecs = eig(K, M)  
  
    return min(eigvals) # Approximation
```

Key Takeaways

- ① **Variational Framework:** Differential equations $\leftrightarrow$ Minimization problems
- ② **Functional Analysis:** Extension of calculus to infinite-dimensional spaces
- ③ **Mountain Pass Lemma:** Geometric principle for finding critical points (solutions)
- ④ **Hamiltonian Systems:** Unifies classical mechanics and quantum mechanics
- ⑤ **Convergence Theory:** Quantifies how fast iterative methods reach fixed points
- ⑥ **Applications:** Physics, mechanics, quantum mechanics, optimization

Connection to Fixed Point Theory

Fixed Point Iteration:

$$x_{n+1} = T(x_n) \Rightarrow x_n \rightarrow x^* \text{ where } T(x^*) = x^*$$

Variational Perspective: Can reformulate as:

$$\min_x \|x - T(x)\|^2$$

Convergence Rate Analysis:

- **Contraction:** Rate depends on contraction constant
- **Non-smooth:** Can use variational methods to analyze rate
- **Quasi-linear:** Intermediate cases between linear and quadratic

Advanced Topic: Modern variational methods use first-order optimization:

- Gradient descent
- Proximal algorithms
- Splitting methods (ADMM, etc.)

Further Reading and Resources

- **Pathak (2018):** “An Introduction to Nonlinear Analysis and Fixed Point Theory”
- **Variational Methods:**
 - Evans, L. C. (1998): “Partial Differential Equations”
 - Dacorogna, B. (2008): “Direct Methods in the Calculus of Variations”
- **Functional Analysis:**
 - Rudin, W. (1991): “Functional Analysis”
 - Brezis, H. (2011): “Functional Analysis, Sobolev Spaces and PDEs”
- **Computational Methods:**
 - Nocedal & Wright (2006): “Numerical Optimization”
 - Boyd & Parikh (2011): “Proximal Algorithms”

Code: Simple Fixed Point Iteration

```
import numpy as np

def fixed_point_iteration(T, x0, tol=1e-10,
                        maxiter=1000):
    x = x0
    errors = [np.linalg.norm(x - T(x))]

    for k in range(maxiter):
        x_new = T(x)
        error = np.linalg.norm(x_new - x)
        errors.append(error)
        if error < tol:
            return x_new, errors
        x = x_new
    return x, errors

# Newton's method for sqrt(2)
def sqrt_iter(x, a=2):
    return 0.5 * (x + a / x)

x_final, errs = fixed_point_iteration(
    lambda x: sqrt_iter(x, 2.0), x0=1.0)
print(f"sqrt(2) = {x_final:.15f}")
```